
Semi-Supervised Approach to Spatial Transcriptomics

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1 Introduction & Problem Definition

1.1 Background

Advances in spatial transcriptomics sequencing technologies have rapidly expanded the scale and complexity of biological data, increasing the need for more powerful computational analysis methods. Spatial transcriptomics (ST) is particularly rich because it measures gene expression while preserving the spatial organization of cells within tissue, enabling the study of cellular neighborhoods and interactions that are lost in standard single-cell RNA-sequencing. These datasets naturally encode relationships between nearby cells, providing a strong opportunity for analysis by graph machine learning. Cells or spots are represented as nodes and the edges are spatial proximity. Graph neural networks (GNNs) are widely used for tasks such as spatial clustering (grouping spots into tissue domains) and cell-type deconvolution.

1.2 Problem Statement

Current GNN-based approaches to ST analysis are predominantly unsupervised, optimizing for clustering quality without leveraging available biological annotations. In a review article analyzing leading GNN architectures for clustering for ST, all of the methods were completely unsupervised (Liu et al., 2024). Thus, when discussing directions of further research, the review suggested integrating multiple methods of learning to improve clustering results, among them semi-supervised methods. It is difficult to simultaneously maximize the clustering ability and the classification ability of a model, as there is generally a trade-off experienced between the two tasks. In this paper, I build off of the GraphST architecture (Long et al., 2023) and incorporate a supervised learning component in the training process, in order to demonstrate the possibility to simultaneously improve clustering and classification.

1.3 Motivation & Significance

ST enables researchers to map gene expression across tissue while preserving spatial context — a capability critical for understanding cell-cell communication, tissue organization, and disease progression. The human cortex, for instance, is organized into six functionally distinct layers whose precise boundaries are essential for studying neurological disorders. Manual annotation of these layers by expert pathologists is expensive, time-consuming, and does not scale to large patient cohorts. Automated cortical layer prediction from ST data would directly accelerate neuroscience research and clinical diagnostics. Most state-of-the-art methods are completely unsupervised models; the goal of this paper is to explore whether incorporating supervised learning in addition to the self-supervised learning can improve performance of the GNN models.

1.4 Learning Objective & Methodology

In my proposed semi-supervised extension of GraphST, there are two objectives: clustering (unsupervised) and classification (supervised). The original GraphST encoder learns to produce spatially coherent embeddings by minimizing the reconstruction and contrastive losses (1). The success of this objective is measured by Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) against ground truth labels after applying KMeans clustering to the embeddings.

With my modification of the loss function, the new model now simultaneously optimizes for clustering and classification (2). The MLP head learns to predict which of the 7 cortical layers each spot belongs to, using the GraphST encoder’s embeddings as input. This is standard node-level classification, with the performance of the classification being measured by Macro-F1 and AUROC. Note that here, while the MLP head learns its classification task using the GraphST encoder’s embeddings, the MLP and GraphST weights are learned and optimized together. This allows the classification loss to flow gradients back through the MLP and into the encoder, shaping the embedding space to be simultaneously spatially coherent and label-discriminative. Meanwhile, in the main baseline, the GraphST encoder produces embeddings, which are then frozen and used to train the MLP classification head (so training and optimization occurs sequentially, rather than simultaneously). This general semi-supervised architecture is motivated by Kipf & Welling (2017) (2).

1.5 Summary

My experiments demonstrate a few key findings. First, GraphST’s self-supervised spatial embeddings are highly predictive of cortical layer identity, even without any supervision during training (Macro-F1 of 0.878 and AUROC of 0.991 after applying MLP head to frozen embeddings). This dramatically outperformed supervised GNNs trained from scratch, as well as the baseline of MLP head applied to data after PCA dimension reduction. Second, joint semi-supervised training not only maintains a similar level of performance with the classification task (or higher, as γ increased), but it also exhibited improved clustering from the original GraphST model. This is a notable result, as it demonstrates that incorporating known labels at train time, in addition the self-supervised contrastive learning, can help the model cluster better at test time. Thus, I found that spatial structure and layer identity are mutually reinforcing signals.

2 Dataset

2.1 Description

For this project, I used a spatial transcriptomics dataset of the Human Dorsolateral prefrontal cortex (DLPFC), obtained through the ST technology of 10x Genomics Visium. The data is organized and available through SpatialLIBD (1). It is used across state-of-the-art architectures, allowing for consistent comparison between my results and other benchmarks.

2.2 Exploratory Data Analysis

Statistic	Value
Number of Spots (Nodes)	4,226
Number of Edges	12,913
Number of Genes	33,538
Graph Density	0.00145
Average Clustering Coefficient	0.4149
Mean Degree	6.11
Min Degree	6
Max Degree	11

Table 1: Graph statistics for DLPFC slice 151673, computed from the k=6 spatial KNN graph.

Note: See appendix for visualization of data and variance techniques performed on dataset.

2.3 Data Splits

I used a single annotated slice (DLPFC slice 151673) which has 4,226 total spots. Of these, 5 lack ground truth annotations and are excluded, leaving 4,221 labeled spots. These are split into 80% training (3,376 spots) and 20% test (845 spots) using stratified sampling to preserve class proportions across the imbalanced cortical layer classes (see figure 3 for visual demonstration of class imbalance). No separate validation set is used during GraphST pretraining, as it is entirely unsupervised. For the MLP classification head, the training split is used for optimization and the test split for final evaluation. For the joint semi-supervised model, the train mask identifies which spots contribute to the classification loss during training (and thus for which spots the model sees the true labels), while the test mask is held out for evaluation (the nodes for which the model predicts the label, but only sees the true label at test time). All splits use random seed 42 for reproducibility.

3 Description of Related Works

I have organized the literature review into three parts: GNN architectures for ST clustering, a comparative benchmark, and a clinical application motivating this project.

STAGATE (3) is a graph attention autoencoder that builds a spatial neighbor network from spot coordinates and learns adaptive edge weights to aggregate expression across neighbors. They show that their introduction of graph attention improves the clustering delineation at domain boundaries. It achieves strong clustering performance on the DLPFC dataset that I am using, and it produces spatially coherent embeddings. However, its objective is purely reconstructive (that is, making the clusters to make the clusters); the model is never trained to predict biological labels.

Meanwhile, **GraphST** (4) instead uses a self-supervised contrastive objective, training a GCN encoder to contrast each spot against a summary of its local spatial neighbors. This yields the strongest clustering embeddings among current methods on the DLPFC benchmark, but like STAGATE, GraphST is entirely unsupervised. The annotations contained in the dataset are never used.

Liu et al. (5) wrote a review benchmarking eight GNN-based ST methods across 60 scenarios (derived from different combinations of GNN architectures, method of downstream clustering, inclusion or exclusion of PCA-based reduction, and method of refinement at the end). They found that GraphST and STAGATE (in addition to Spatial-MGCN, a third paper not discussed in this literature review) achieved the best clustering performance over the eight benchmark methods Liu et al. evaluated on. Critically, none of the eight methods attempt to use their learned embeddings for supervised label prediction, but rather just measure the accuracy of the clustering itself.

Finally, **NePSTA** (6) shows what is possible when spatial transcriptomics is analyzed with a supervised GNN. They train a GIN on 3-hop Visium subgraphs from 130 CNS tumor patients to predict methylation-based tumor subclasses, which performs incredibly well. However, NePSTA is trained end-to-end from scratch, with no spatial pretraining. This project asks whether initializing from GraphST’s pretrained embeddings could improve such supervised classifiers.

4 Methodology

4.1 GraphST Baseline

My work builds directly on GraphST (Long et al., 2023), which I use as both a baseline and the foundation for my novel extension. GraphST constructs a spatial neighbor graph from tissue coordinates, connecting each spot to its $k = 6$ nearest neighbors. A two-layer Graph Convolutional Network (GCN) encoder maps each spot’s normalized gene expression vector (3,000 highly variable genes selected via `seurat_v3` flavor) to a 64-dimensional latent embedding $z_i \in \mathbb{R}^{64}$. The encoder is trained by minimizing the self-supervised loss described in their paper:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{recon} + \lambda_2 (\mathcal{L}_{SCL} + \mathcal{L}_{SCL_{corrupt}}) \quad (1)$$

where $\mathcal{L}_{recon}$ is an MSE reconstruction loss encouraging the encoder to preserve gene expression information, $\mathcal{L}_{SCL}$ is a spatial contrastive loss on the real graph that pushes spatially adjacent spots together and non-adjacent spots apart, and $\mathcal{L}_{SCL_{corrupt}}$ applies the same contrastive objective on a

corrupted graph where node features are randomly permuted across spots. Following the parameters used in Long et al., I set $\lambda_1 = 10$, $\lambda_2 = 1$, and learning rate = 0.001, training for 600 epochs. (Note that in the code, λ_1 and λ_2 are referred to as alpha and beta, respectively, for ease of referencing.) All GraphST code for this initial section is used directly from the paper’s github repository without modification.

I chose GraphST as the base architecture given the findings in the review by Liu et al. (2024) that GraphST was the best overall model for ST clustering analysis.

4.2 Frozen GraphST -> MLP

To create a strong baseline for my novel contribution described in 4.3, I first evaluate whether GraphST’s self-supervised embeddings are predictive of cortical layer identity without any modification to GraphST’s training procedure. After training GraphST as described above, the encoder weights are frozen and the 64-dimensional embeddings are extracted for all 4,221 labeled spots. A classic two-layer MLP classifier is then trained on the 80% training split with hidden dimensions 256 and 128 (dimensions go from $64 \rightarrow 256 \rightarrow 128 \rightarrow 7$), and then each spot is classified based on its embedding output, representing probabilities of the spot’s true label.

4.3 Joint Semi-Supervised GraphST

My key contribution is modification of GraphST’s training objective to incorporate a supervised classification signal directly into the pretraining loop. Rather than training GraphST unsupervised and then applying a classifier to frozen embeddings, I jointly optimized the encoder and a classification head end-to-end using a combined loss:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{recon} + \lambda_2 (\mathcal{L}_{SCL} + \mathcal{L}_{SCL_{corrupt}}) + \gamma \mathcal{L}_{cls} \quad (2)$$

where $\mathcal{L}_{cls}$ is a class-weighted cross-entropy loss computed only on the labeled training spots, and $\gamma \geq 0$ is a novel hyperparameter that controls the amount of supervision. When $\gamma = 0$, my model reduces to the original GraphST loss function. When $\gamma > 0$, the encoder simultaneously learns spatially coherent and label-discriminative representations. In particular, gradients from the classification loss flow back through the MLP head and into the GCN encoder, shaping the 64-dimensional embedding space to perform well in both clustering and classification tasks.

The architecture consists of the original GraphST GCN encoder (unchanged) combined with a two-layer MLP classification head identical to the one described in Section 4.2. All parameters — encoder and classifier — are optimized jointly using a single Adam optimizer (lr=0.001) for 600 epochs. The unsupervised losses use the same hyperparameters as the original GraphST ($\lambda_1 = 10$ and $\lambda_2 = 1$). I ablate $\gamma \in 0, 0.1, 0.5, 1.0, 5.0$ to characterize the tradeoff of greater supervision. This modification directly addresses the future research direction identified by Liu et al. (2024) by integrating contrastive and semi-supervised methods into spatial clustering frameworks. In the spatial transcriptomics setting, joint training is particularly motivated by the fact that cortical layers are spatially contiguous, meaning the spatial contrastive signal and the layer classification signal are mutually reinforcing rather than conflicting objectives.

My novel contributions include:

1. The addition of $\gamma \mathcal{L}_{cls}$ to the training objective
2. The JointGraphST PyTorch class combining the GraphST encoder with an MLP classification head
3. The `run_joint_graphst` training function implementing the combined loss; and
4. The γ ablation study characterizing the supervision strength tradeoff.

All code is available at https://github.com/YossiM11/ST_GNN_Project/tree/main.

Model	Macro-F1	AUROC	ARI	NMI
Raw counts $\rightarrow$ MLP	0.5606	0.8957	—	—
PCA $\rightarrow$ MLP	0.6211	0.9270	—	—
Supervised GCN (scratch)	0.1486	0.5041	—	—
Supervised GIN (scratch)	0.1462	0.4908	—	—
Frozen GraphST $\rightarrow$ MLP	0.8784	0.9905	0.4162	0.5322
Joint GraphST ($\gamma = 0$)	0.1055	0.4908	0.4162	0.5322
Joint GraphST ($\gamma = 0.1$)	0.8174	0.9697	0.3900	0.5316
Joint GraphST ($\gamma = 0.5$)	0.8504	0.9776	0.4549	0.5641
Joint GraphST ($\gamma = 1.0$)	0.8727	0.9822	0.4558	0.5664
Joint GraphST ($\gamma = 5.0$)	0.9100	0.9885	0.4542	0.5657

Table 2: Results on DLPFC slice 151673 (80/20 stratified split).

5 Experiments & Results

5.1 Experimental Setup

5.1.1 Baselines

I compared my model against: Raw counts $\rightarrow$ MLP; PCA $\rightarrow$ MLP; Supervised GCN from scratch; Supervised GIN from scratch; and Frozen GraphST $\rightarrow$ MLP. The supervised GCN and GIN baselines test whether supervised GNNs without the GraphST training can match my approach. (Note: the code for GCN and GIN is not in the github repository. I only report the results, given that none of it is original code, but just a classic implementation of GCN and GIN.)

5.1.2 Hyperparameters

For the GraphST architecture, I used the original paper’s hyperparameters ($\lambda_1 = 10$, $\lambda_2 = 1$, lr = 0.001, 600 epochs). The MLP goes from 64 $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 7 (thus 256 and 128 are hidden dimensions) and uses dropout of $p = 0.3$, lr=0.001, and 300 epochs. My novel γ parameters is ablated across 0, 0.1, 0.5, 1.0, 5.0. All experiments use random seed 42.

5.1.3 Computational details

The training was performed locally on CPU. GraphST training alone took 3.5 minutes. Joint GraphST took 4.5 minutes per γ . MLP training took 1.5 minutes.

5.2 Quantitative Results

I use **Macro-F1** and **AUROC** to report classification performance, weighting all seven classes equally to account for significant class imbalance (e.g. Layer 3 has 1,215 spots vs. Layer 2 has 305 spots). To evaluate the clustering task, I report **ARI** and **NMI**, which is consistent with the GraphST benchmark literature (Liu et al., 2024).

5.3 Analysis & Discussion

Three main findings emerge from Table 2. First, supervised GNNs trained from scratch perform worse than a simple MLP on raw counts, confirming that graph structure without spatial pretraining or simple dimension reduction is harmful. Second, frozen GraphST embeddings achieve F1 = 0.878 and AUROC = 0.991 without any label supervision during training, validating my core hypothesis that self-supervised spatial pretraining produces highly label-discriminative representations. Third, joint semi-supervised training ($\gamma = 5.0$) further improves classification according to the Macro-F1 metric (F1 = 0.910), which is particularly meaningful here, given that Macro-F1 specializes in assessing classification under class imbalance, which is indeed the case. Notably, moderate supervision improves the clustering quality (e.g. ARI increases from frozen baseline of 0.416 to 0.456 at $\gamma = 1.0$), while performing quite well in classification tasks as well.

The γ ablation (observable in Image 4) reveals that even minimal supervision ($\gamma = 0.1$) unlocks strong classification performance. Interestingly, at $\gamma = 0.1$, the clustering performance is slightly worse than the frozen GraphST baseline, but by $\gamma = 0.5$, the joint model is outperforming the frozen

GraphST baseline. We find that classification and clustering performance improves from $\gamma = 0.5$ to $\gamma = 1.0$, and then classification performance continues to improve at $\gamma = 5.0$, while clustering performance decreases slightly from $\gamma = 1.0$. This trend suggests that the proper balance between optimizing for classification and clustering lies somewhere between $\gamma = 1.0$ and $\gamma = 5.0$.

Finally, note that for the Joint GraphST with $\gamma = 0$, the classification performance is poor, given that it has a completely untrained classifier (classification is not accounted for at all, neither in the loss function nor in any downstream application as it is in the Frozen GraphST to MLP baseline).

One weakness of this study was that it only examined one slice (i.e. dataset) from the DLPFC library. Though this was also what the GraphST paper did for most of its tests, future research should confirm that these results uphold for the other slices. More importantly, future research should utilize Leave-One-Out Cross-Validation so that training can be done on entire slices, and then testing can examine whether the embeddings generalize well to entirely new slices, rather than just different spots within the same slice (as I measured here). I attempted to explore LOOCV but faced many difficulties with different embedding structures across different slices, so future research in this area would be beneficial. This would also be particularly helpful when applying the findings to the medical field, where ST slices are completely unannotated, so if this joint model can generalize well to new slices, then it can be trained on pre-labeled slices (as in DLPFC) and then be applied to new patient ST readings.

An additional weakness to note is that the ARI and NMI reported for the GraphST is significantly lower than the original GraphST reported (theirs was around 0.6, rather than 0.4). This can be explained by the fact that I used simpler clustering techniques once the embeddings were generated, and had less post-processing to remove noise. Fortunately, this is not problematic within the scope of this paper, given that the key finding stands: the joint GraphST model outperforms the original GraphST in clustering at certain gamma values. So, this finding can be expected to apply also when there is better post-processing.

Returning to my original research question, I had asked: "Does using GraphST's spatially-pretrained embeddings as a starting point improve cortical layer classification compared to a supervised GNN trained from scratch?" I hypothesized, "Yes, using GraphST as pretraining will allow my model to use embeddings that better capture the spatial tissue organization, which will in turn allow the additional supervision learning MLP stage to outperform a generic supervised GNN classification model that is operating without GraphST pretraining." My findings affirm my original hypothesis, and also go beyond it with additional novel findings. I answered my original research question through the Frozen GraphST to MLP baseline, demonstrating that it outperformed all other baselines in classification. I additionally found that incorporating the MLP head into the training process itself, optimizing it alongside the GraphST encoder, improves both classification performance as well as clustering performance.

6 Conclusion

I presented a semi-supervised extension of GraphST that jointly optimizes spatial contrastive pre-training and cortical layer classification through a combined loss function. My experiments on DLPFC slice 151673 demonstrate that GraphST's self-supervised spatial embeddings, when frozen and an MLP classification head is applied, are highly predictive of cortical layer identity, dramatically outperforming other baselines. Additionally, joint semi-supervised training can improve the clustering performance above the GraphST baseline, while maintaining a similar (and sometimes improved, in the case of $\gamma = 5.0$) classification performance to the frozen GraphST embeddings. This novel semi-supervised architecture combined the strong clustering performance of GraphST with the call to further research in semi-supervised methods in Liu et al. (2024).

Some of the main lessons learned include that embeddings don't necessarily align across different slices, hence causing issues in LOOCV when trying to combine embeddings that were trained independently, on different slices. I also ran into technical difficulties regarding the Python edition (e.g. edition required for scikit-misc download vs edition GraphST was built around vs edition the Yale cluster runs on), and I had to work on the pre-processing of the datasets so that the inputs to GraphST matched what the paper expected. Limitations, as discussed above, are the fact that training was performed on one slice only. Future work should apply methods such as PASTE (7), which GraphST applies, to this novel Joint semi-supervised GraphST model.

References

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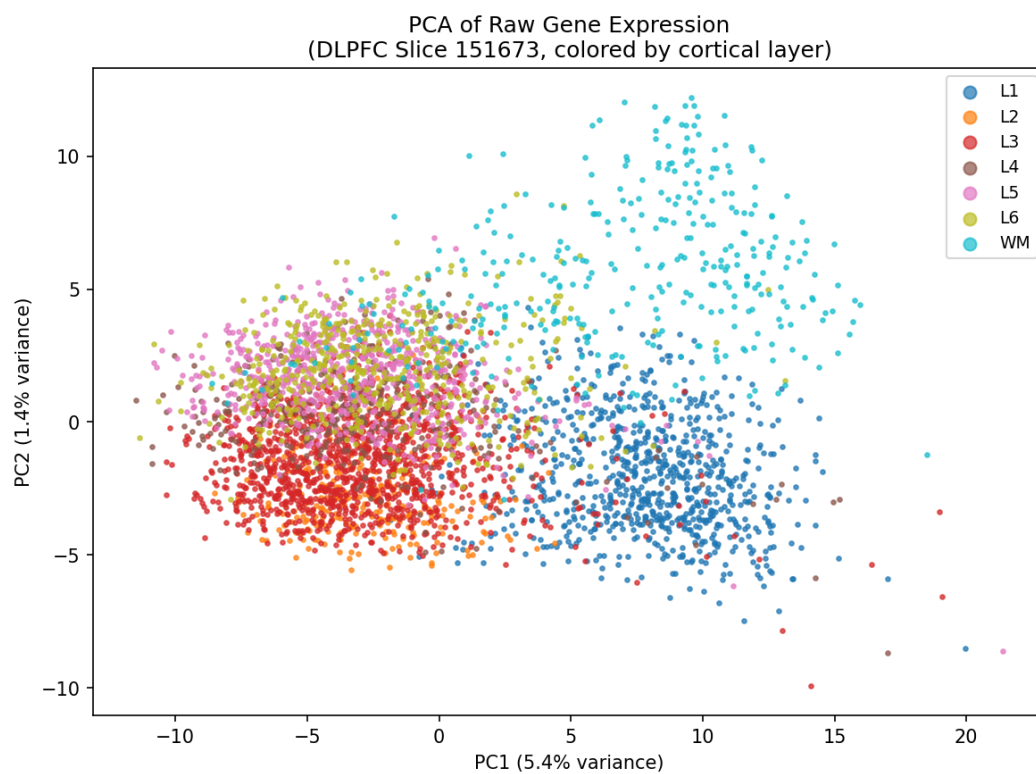


Figure 1: PCA dimension reduction performed on initial 151673 dataset, colored by layer. It only captures 6.8% of variance in data between PC1 and PC2.



Figure 2: t-SNE of 151673 data, colored by layer.

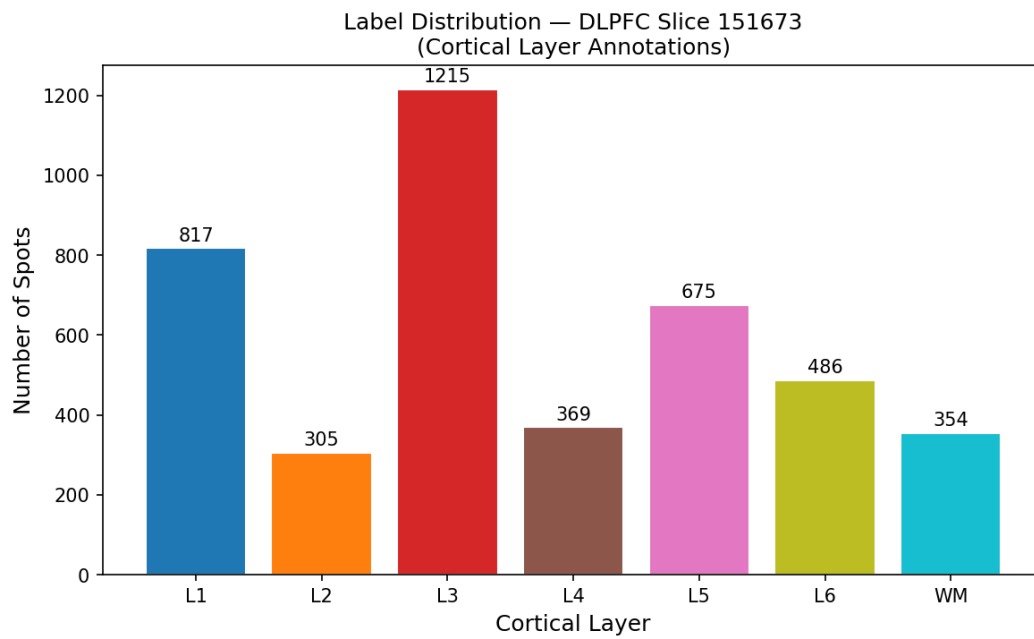


Figure 3: Label distribution of dataset 151673. Note the class imbalance (up to 4x difference between largest and smallest class). This is accounted for in the training.

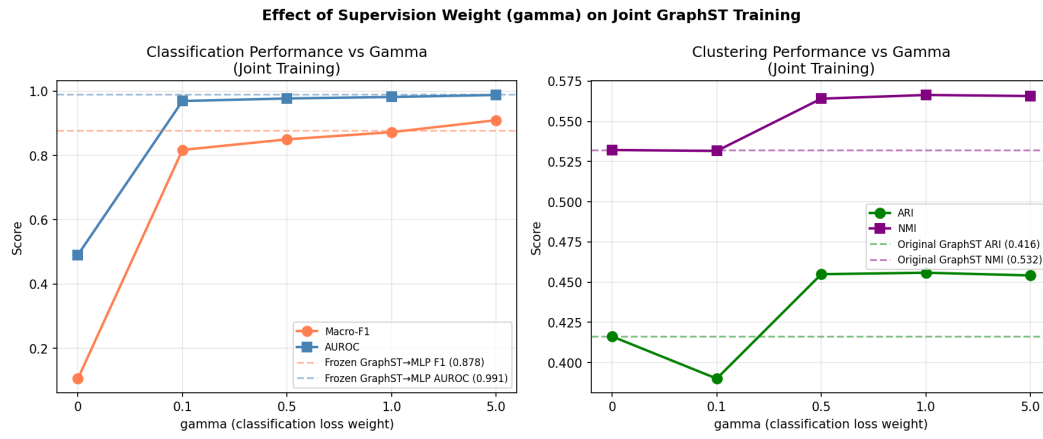


Figure 4: Classification and Clustering Performance of Joint GraphST at varying γ values.